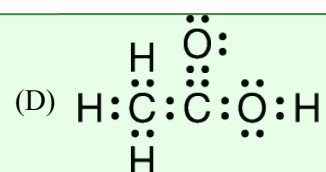
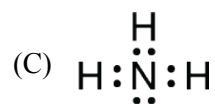
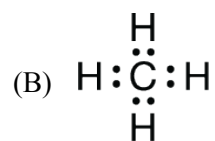
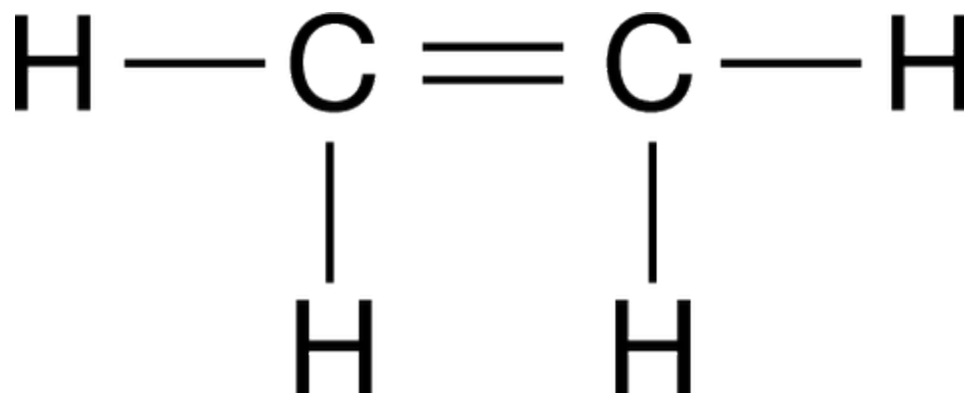


Chapter 02 Part 2: Molecular Structures

1. Which of the following complete Lewis diagrams represents a molecule containing a bond angle that is closest to 120° ?



2.



A Lewis diagram for the molecule C_2H_4 is shown above. In the actual C_2H_4 molecule, the $\text{H}-\text{C}-\text{H}$ bond angles are closest to

- (A) 90°
(B) 109.5°

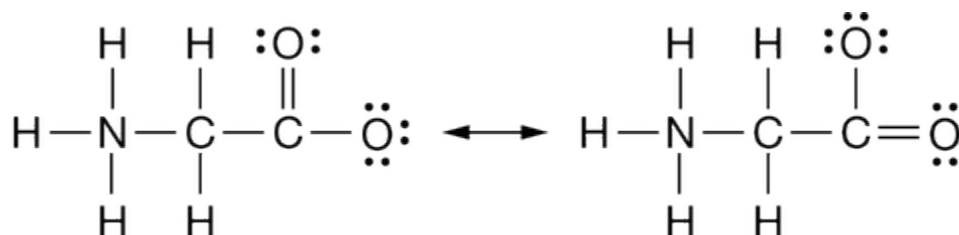
(C) 120°

(D) 180°



Chapter 02 Part 2: Molecular Structures

3.



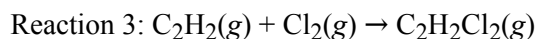
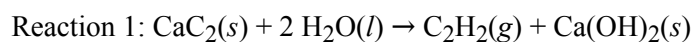
Based on the resonance structures shown above, what are the bond orders of the two carbon-oxygen bonds?

(A) 1 and 1.5

(B) 1 and 2

(C) 1.5 and 1.5

(D) 2 and 2



4. When Reaction 3 occurs, does the hybridization of the carbon atoms change?

(A) Yes; it changes from sp to sp^2 .(B) Yes; it changes from sp to sp^3 .(C) Yes; it changes from sp^2 to sp^3 .

(D) No; it does not change



5.



Lewis electron-dot diagrams for CO_2 and SO_2 are given above. The molecular geometry and polarity of the two substances are

(A) the same because the molecular formulas are similar

(B) the same because C and S have similar electronegativity values

(C) different because the lone pair of electrons on the S atom make it the negative end of a dipole

(D) different because S has a greater number of electron domains (regions of electron density) surrounding it than C has



Chapter 02 Part 2: Molecular Structures

Refer to the following diatomic species

(A) Li_2

(B) B_2

(C) N_2

(D) O_2

(E) F_2

6. Contains 1 sigma (σ) and 2 pi (π) bonds

(A) Li_2

(B) B_2

(C) N_2

(D) O_2

(E) F_2



7. Has a bond order of 2

(A) Li_2

(B) B_2

(C) N_2

(D) O_2

(E) F_2



8. Has the largest bond-dissociation energy

(A) Li_2

(B) B_2

(C) N_2

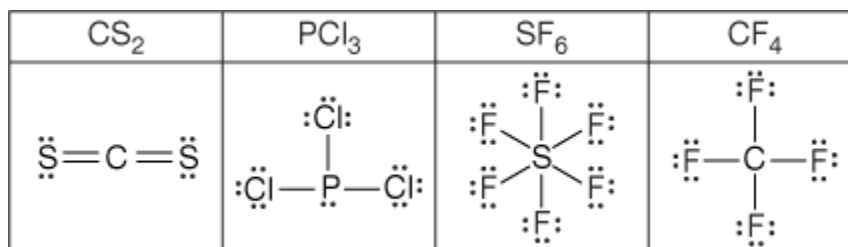
(D) O_2

(E) F_2



Chapter 02 Part 2: Molecular Structures

9.



Based on the diagrams shown above, which of the following would have the greatest dipole moment, and why?

- (A) CS₂, because the double bonds create an area of higher electron density around the C atom.
- (B) PCl₃, because the Cl atoms draw electron density away from the P atom and the bond dipoles do not cancel one another. ✓
- (C) SF₆, because it has the greatest number of polar bonds.
- (D) CF₄, because the large electronegativity difference between C and F results in very polar bonds.

Directions: Each set of lettered choices below refers to the numbered statement immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

Refer to the following gaseous molecules:

- (A) BeCl₂
- (B) SO₂
- (C) N₂
- (D) O₂
- (E) F₂

10. Is best represented by two or more resonance forms

- (A) BeCl₂
- (B) SO₂ ✓
- (C) N₂
- (D) O₂
- (E) F₂

11. Is the molecule in which the intramolecular forces are strongest

Chapter 02 Part 2: Molecular Structures(A) BeCl_2 (B) SO_2 (C) N_2 (D) O_2 (E) F_2 **12.** Is a polar molecule(A) BeCl_2 (B) SO_2 (C) N_2 (D) O_2 (E) F_2 **13.** The geometry of the SO_3 molecule is best described as

(A) trigonal planar



(B) trigonal pyramidal

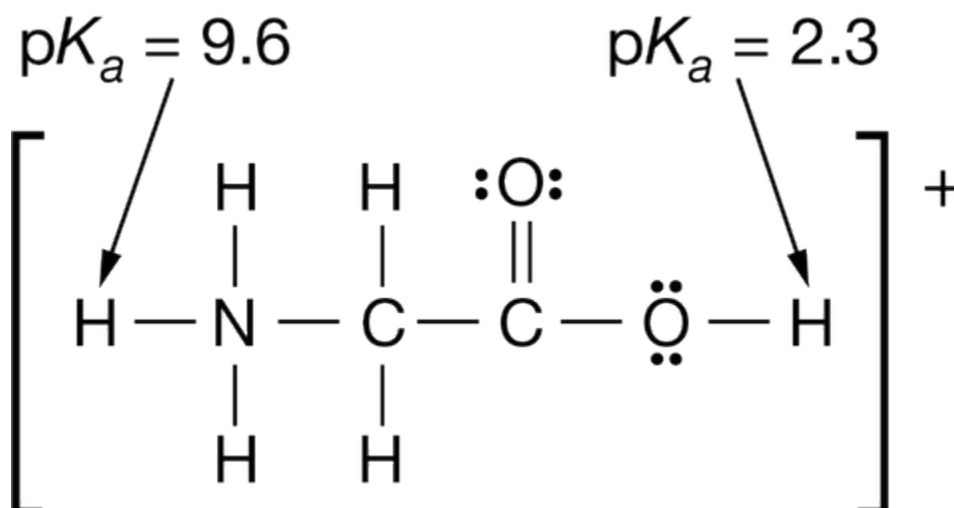
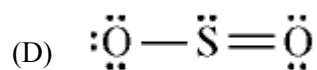
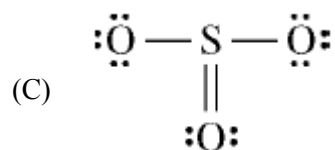
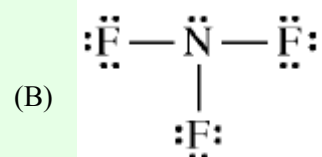
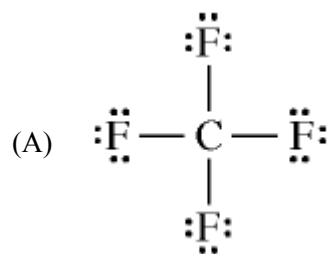
(C) square pyramidal

(D) bent

(E) tetrahedral

14. Which of the following Lewis electron-dot diagrams represents the molecule that contains the smallest bond angle?

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The structural formula of the glycinium cation is shown above. Arrows indicate the $\text{p}K_a$ values for the labile protons in the molecule.

15. Which of the following is true about the geometry of the glycinium cation?

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(A) The leftmost C atom and all the atoms directly bonded to it lie in the same plane.

(B) Both C atoms and both O atoms lie in the same plane. ✓

(C) The N – C – C bond angle is 180° .

(D) The geometry around the N atom is planar.

16. What is the approximate H – O – C bond angle in the glycinium cation?

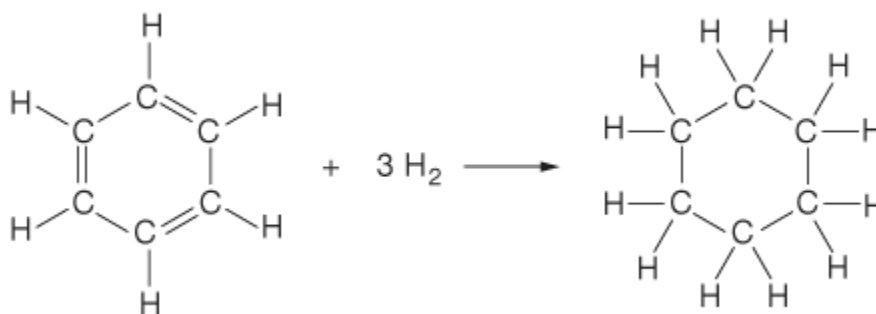
(A) 180°

(B) 120°

(C) 105° ✓

(D) 90°

17.



In the reaction represented above, which of the following sets correctly identifies the hybridization of the C atoms before and after the reaction occurs?

Set

Before

After

sp^2 sp^2

A

sp^2 sp^3

B

sp^3 sp^2

C

sp^3 sp^3

D

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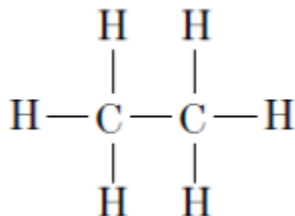
(A) Set A

(B) Set B

(C) Set C

(D) Set D

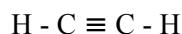
18.



The hybridization of the carbon atoms in the molecule represented above can be described as

(A) sp (B) sp^2 (C) sp^3 (D) dsp^2 (E) d^2sp

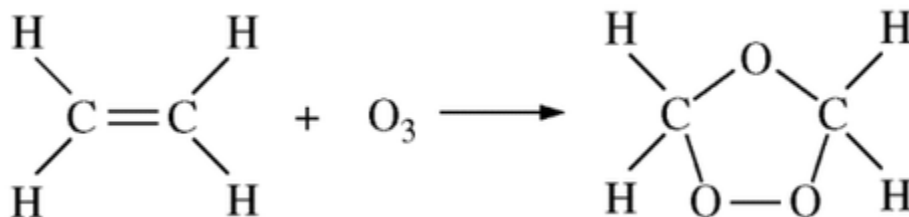
19.



What is the hybridization of the carbon atoms in a molecule of ethyne, represented above?

(A) sp (B) sp^2 (C) sp^3 (D) dsp^2 (E) d^2sp

20.



In the reaction represented above, what is the hybridization of the C atoms before and after the reaction occurs?

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(A)	<u>Before</u>	<u>After</u>
	sp	sp^2

(B)	<u>Before</u>	<u>After</u>
	sp	sp^3

(C)	<u>Before</u>	<u>After</u>
	sp^2	sp

(D)	<u>Before</u>	<u>After</u>	
	sp^2	sp^3	

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

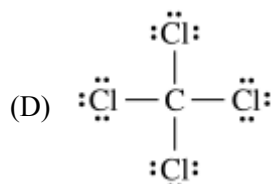
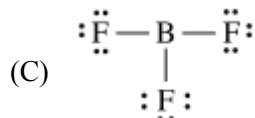
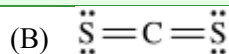
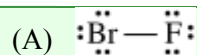
The question refers to the following species.

- (A) H_2O
- (B) NH_3
- (C) BH_3
- (D) CH_4
- (E) SiH_4

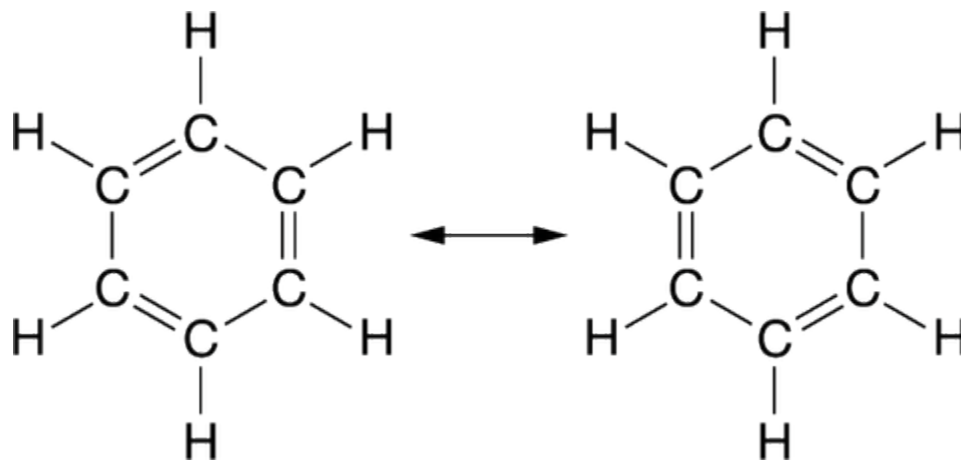
21. Has a central atom with less than an octet of electrons

Chapter 02 Part 2: Molecular Structures(A) H_2O (B) NH_3 (C) BH_3 ✓(D) CH_4 (E) SiH_4 **22.** Is predicted to have the largest bond angle(A) H_2O (B) NH_3 (C) BH_3 ✓(D) CH_4 (E) SiH_4 **23.** Has a trigonal-pyramidal molecular geometry(A) H_2O (B) NH_3 ✓(C) BH_3 (D) CH_4 (E) SiH_4 **24.** Has two lone pairs of electrons(A) H_2O ✓(B) NH_3 (C) BH_3 (D) CH_4 (E) SiH_4 **25.** Which of the following molecules has the largest dipole moment?

Chapter 02 Part 2: Molecular Structures



26.



The diagram above shows two resonance structures for a molecule of C_6H_6 . The phenomenon shown in the diagram best supports which of the following claims about the bonding in C_6H_6 ?

(A) In the C_6H_6 molecule, all the bonds between the carbon atoms have the same length.

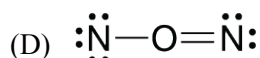
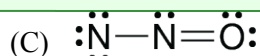
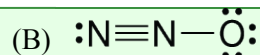
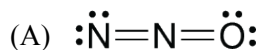


(B) Because of variable bonding between its carbon atoms, C_6H_6 is a good conductor of electricity.

(C) The bonds between carbon atoms in C_6H_6 are unstable, and the compound decomposes quickly.

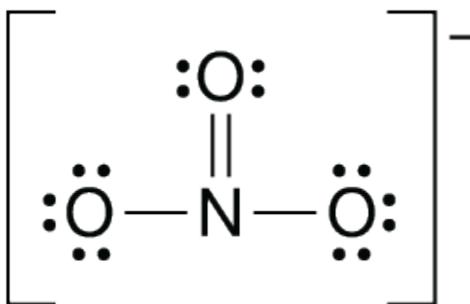
(D) The C_6H_6 molecule contains three single bonds between carbon atoms and three double bonds between carbon atoms.

27. Which of the following Lewis diagrams best represents the bonding in the N_2O molecule, considering formal charges?



Chapter 02 Part 2: Molecular Structures

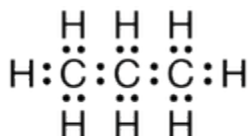
28.



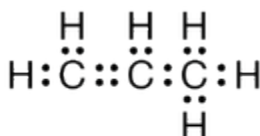
Which of the following statements, if true, would support the claim that the NO_3^- ion, represented above, has three resonance structures?

- (A) The NO_3^- ion is not a polar species. ✓
- (B) The oxygen-to-nitrogen-to-oxygen bond angles are 90° .
- (C) One of the bonds in NO_3^- is longer than the other two.
- (D) One of the bonds in NO_3^- is shorter than the other two.

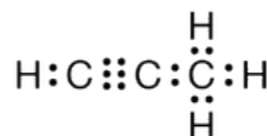
29.



Molecule 1



Molecule 2



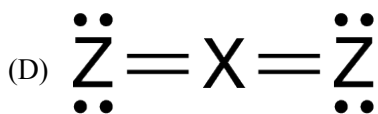
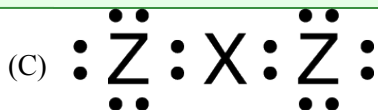
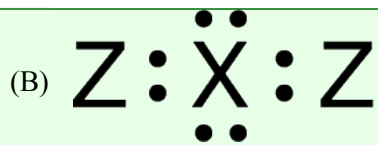
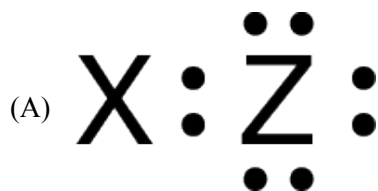
Molecule 3

Lewis diagrams of molecules of three different hydrocarbons are shown above. Which of the following claims about the molecules is best supported by the diagrams?

- (A) All the atoms in molecule 1 lie in one plane.
- (B) All the molecules have the same empirical formula.
- (C) The C-C-C bond angle in molecule 2 is close to 180° .
- (D) The strongest carbon-to-carbon bond occurs in molecule 3. ✓

30. In the following diagrams, elements are represented by X and Z, which form molecular compounds with one another. Which diagram represents a molecule that has a bent molecular geometry?

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Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

The following questions refer to the below molecules:

(A) CO_2

(B) H_2O

(C) CH_4

(D) C_2H_4

(E) PH_3

31. The molecule with the largest dipole moment

(A) CO_2

(B) H_2O ✓

(C) CH_4

(D) C_2H_4

(E) PH_3

32. The molecule with only one double bond

(A) CO_2

(B) H_2O

(C) CH_4

(D) C_2H_4 ✓

(E) PH_3

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33. The molecule that has trigonal pyramidal geometry

- (A) CO_2
- (B) H_2O
- (C) CH_4
- (D) C_2H_4

(E) PH_3



34. For which of the following molecules are resonance structures necessary to describe the bonding satisfactorily?

(A) H_2S

(B) SO_2



(C) CO_2

(D) OF_2

(E) PF_3

35. Which of the following molecules contains exactly three sigma (σ) bonds and two pi (π) bonds?

(A) C_2H_2



(B) CO_2

(C) HCN

(D) SO_3

(E) N_2

36. Which of the following molecules has an angular (bent) geometry that is commonly represented as a resonance hybrid of two or more electron-dot structures?

(A) CO_2

(B) O_3



(C) CH_4

(D) BeF_2

(E) OF_2

37. Of the following molecules, which has the largest dipole moment?

(A) CO

(B) CO_2

(C) O_2

(D) HF



(E) F_2

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38. Has molecules with a pyramidal shape

(A) $\text{NH}_3(\text{g})$

(B) $\text{BH}_3(\text{g})$

(C) $\text{H}_2(\text{g})$

(D) $\text{H}_2\text{S}(\text{g})$

(E) $\text{HBr}(\text{g})$

39. Which of the following Lewis electron-dot diagrams represents the molecule that is the most polar?

(A) $\text{:}\ddot{\text{Cl}}-\ddot{\text{F}}\text{:}$

(B) $\text{:}\ddot{\text{Br}}-\ddot{\text{F}}\text{:}$

(C) $\text{:}\ddot{\text{S}}=\text{C}=\ddot{\text{S}}\text{:}$

(D) $\begin{array}{c} \text{:}\ddot{\text{F}}-\text{B}-\ddot{\text{F}}\text{:} \\ | \\ \text{:}\ddot{\text{F}}\text{:} \end{array}$

40. Which of the following molecules contains polar covalent bonds but is a nonpolar molecule?

(A) CH_3Cl

(B) CH_2Cl_2

(C) NH_3

(D) CCl_4

(E) N_2

41. Which of the following molecules is nonpolar but has polar covalent bonds?

(A) N_2

(B) H_2O_2

(C) H_2O

(D) CCl_4

(E) CH_2Cl_2

42. Which of the following arranges the molecules N_2 , O_2 , and F_2 in order of their bond enthalpies, from least to greatest?

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(A) $F_2 < O_2 < N_2$



(B) $O_2 < N_2 < F_2$

(C) $N_2 < O_2 < F_2$

(D) $N_2 < F_2 < O_2$

43. Pi (π) bonding occurs in each of the following species EXCEPT

(A) CO_2

(B) C_2H_4

(C) CN^-

(D) C_6H_6

(E) CH_4



44. Resonance is most commonly used to describe the bonding in molecules of which of the following?

(A) CO_2

(B) O_3



(C) H_2O

(D) CH_4

(E) SF_6

45. Which of the following molecules has the shortest bond length?

(A) N_2



(B) O_2

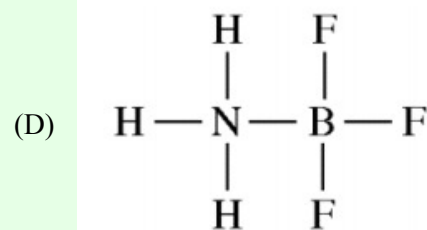
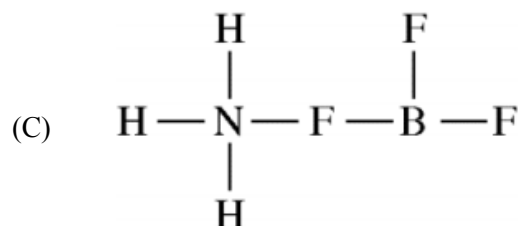
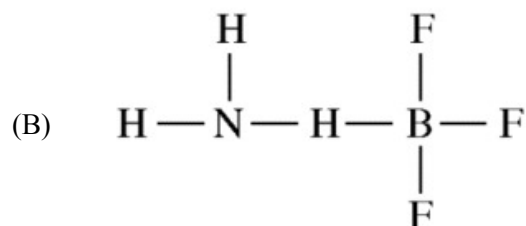
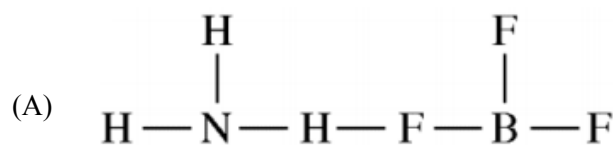
(C) Cl_2

(D) Br_2

(E) I_2

46. NH_3 reacts with BF_3 to form a single species. Which of the following structural diagrams is the most likely representation of the product of the reaction?

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47. According to the VSEPR model, the progressive decrease in the bond angles in the series of molecules CH_4 , NH_3 , and H_2O is best accounted for by the
- (A) increasing strength of the bonds
 - (B) decreasing size of the central atom
 - (C) increasing electronegativity of the central atom
 - (D) increasing number of unshared pairs of electrons
 - (E) decreasing repulsion between hydrogen atoms
48. The electron-dot structure (Lewis structure) for which of the following molecules would have two unshared pairs of electrons on the central atom?

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(A) H_2S (B) NH_3 (C) CH_4 (D) HCN (E) CO_2

49. Which of the following is a nonpolar molecule that contains polar bonds?

(A) F_2 (B) CHF_3 (C) CO_2 (D) HCl (E) NH_3 50. Which of the molecules represented below contains carbon with sp^2 hybridization?(A) CH_4 (B) CH_2Cl_2 (C) C_2H_6 (D) $\text{C}_2\text{H}_2\text{Cl}_2$ (E) $\text{C}_2\text{H}_4\text{Cl}_2$

51. Which of the following molecules contains only single bonds?

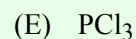
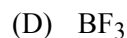
(A) CH_3COOH (B) $\text{CH}_3\text{CH}_2\text{COOCH}_3$ (C) C_2H_6 (D) C_6H_6 (E) HCN

52. Which of the following has a zero dipole moment?

(A) HCN (B) NH_3 (C) SO_2 (D) NO_2 (E) PF_5 

53. Which of the following species is NOT planar?

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54. The BF_3 molecule is nonpolar, whereas the NF_3 molecule is polar. Which of the following statements accounts for the difference in polarity of the two molecules?

(A) In NF_3 , each F is joined to N with multiple bonds, whereas in BF_3 , each F is joined to B with single bonds.

(B) N — F bonds are polar, whereas B — F bonds are nonpolar.

(C) NF_3 is an ionic compound, whereas BF_3 is a molecular compound.

(D) Unlike BF_3 , NF_3 has a nonplanar geometry due to an unshared pair of electrons on the N atom. ✓

55. Account for each of the following observations in terms of atomic theory and/or quantum theory.

a. Atomic size decreases from Na to Cl in the periodic table.

b. Boron commonly forms molecules of the type BX_3 . These molecules have a trigonal planar structure.

c. The first ionization energy of K is less than that of Na.

d. Each element displays a unique gas-phase emission spectrum.

Part A

1 point is earned for indicating the increase in nuclear charge.

1 point is earned for attributing the size decrease to the greater attraction of the nucleus for the electrons caused by the increase in nuclear charge.

Across the periodic table from Na to Cl, the number of electrons in the *s*- and *p*- orbitals of the valence shell increases, as does the number of protons in the nucleus. The added electrons only partially shield the added protons, resulting in an increased effective nuclear charge. This results in a greater attraction for the electrons, drawing them closer to the nucleus, making the atom smaller.

✓

0	1	2
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The student response earns two of the following points:

1 point is earned for indicating the increase in nuclear charge.

1 point is earned for attributing the size decrease to the greater attraction of the nucleus for the electrons caused by the

Chapter 02 Part 2: Molecular Structures

increase in nuclear charge.

Across the periodic table from Na to Cl, the number of electrons in the *s*- and *p*- orbitals of the valence shell increases, as does the number of protons in the nucleus. The added electrons only partially shield the added protons, resulting in an increased effective nuclear charge. This results in a greater attraction for the electrons, drawing them closer to the nucleus, making the atom smaller.

Part B

1 point is earned for describing the valence electrons and the bonds.

1 point is earned for a correct VSEPR argument.

Boron has three valence electrons, each of which can form a single covalent bond with X. The three single covalent bonds of the boron atom orient to minimize electron-pair interaction, resulting in bond angles of 120° and a trigonal planar structure.



0	1	2
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The student response earns two of the following points:

1 point is earned for describing the valence electrons and the bonds.

1 point is earned for a correct VSEPR argument.

Boron has three valence electrons, each of which can form a single covalent bond with X. The three single covalent bonds of the boron atom orient to minimize electron-pair interaction, resulting in bond angles of 120° and a trigonal planar structure.

Part C

1 point is earned for the size explanation.

1 point is earned for describing the attraction to the nucleus.

Both Na and K have an *s*1 valence-shell electron configuration (Na: [Ne] 3*s*1 ; K: [Ar] 4*s*1). The K atom valence electron has a higher *n* quantum number, placing it farther from the nucleus than the Na atom valence electron. The greater distance results in less attraction to the nucleus. Because its valence electron is less attracted to its nucleus, the K atom has the lower ionization energy.



0	1	2
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The student response earns two of the following points:

1 point is earned for the size explanation.

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1 point is earned for describing the attraction to the nucleus.

Both Na and K have an $s1$ valence-shell electron configuration (Na: $[\text{Ne}] 3s1$; K: $[\text{Ar}] 4s1$). The K atom valence electron has a higher n quantum number, placing it farther from the nucleus than the Na atom valence electron. The greater distance results in less attraction to the nucleus. Because its valence electron is less attracted to its nucleus, the K atom has the lower ionization energy.

Part D

1 point is earned for describing the quantized energy states and emission phenomenon.

1 point is earned for describing the effect of the uniqueness of the nucleus and/or electron configuration.

Each element has a unique set of quantized energy states for its electrons (because of its unique nuclear charge and unique electron configuration). As the electrons of the element absorb quanta of energy, they change to higher energy states (are excited) – during de-excitation, energy is released as EM radiation as the electrons cascade to lower energy states. Each photon of the EM radiation is associated with a specific wavelength ($\lambda = hc/E$), a flux of which produces the lines of the emission spectrum.



0	1	2
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The student response earns two of the following points:

1 point is earned for describing the quantized energy states and emission phenomenon.

1 point is earned for describing the effect of the uniqueness of the nucleus and/or electron configuration.

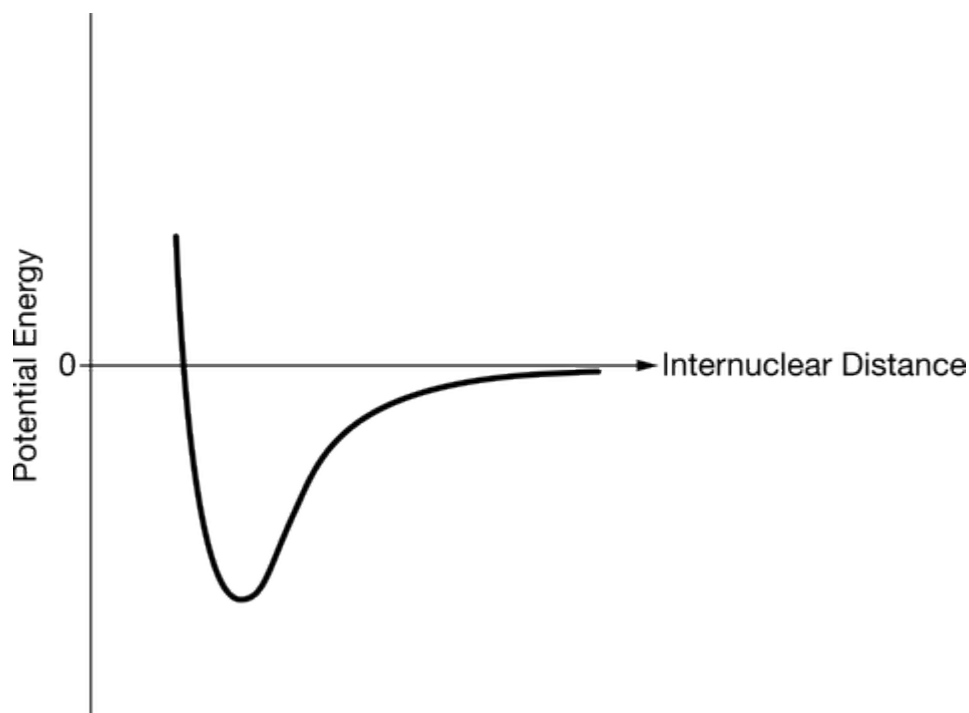
Each element has a unique set of quantized energy states for its electrons (because of its unique nuclear charge and unique electron configuration). As the electrons of the element absorb quanta of energy, they change to higher energy states (are excited) – during de-excitation, energy is released as EM radiation as the electrons cascade to lower energy states. Each photon of the EM radiation is associated with a specific wavelength ($\lambda = hc/E$), a flux of which produces the lines of the emission spectrum.

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56. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

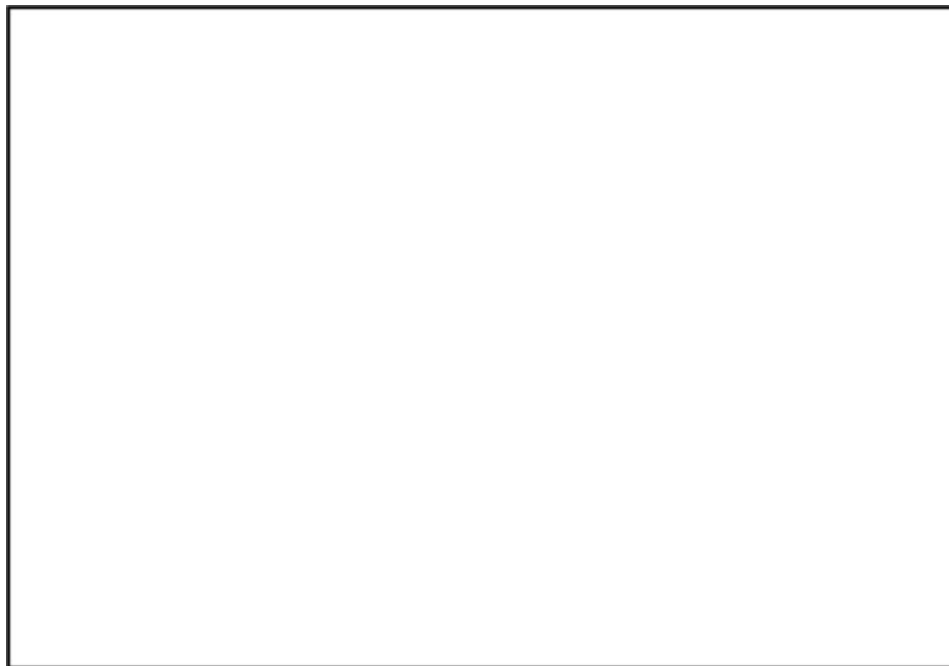
Answer the following questions related to the chemical bonding in substances containing Cl.

- (a) What type of chemical bond is present in the Cl_2 molecule?
- (b) Cl_2 reacts with the element Sr to form an ionic compound. Based on periodic properties, identify a molecule, X_2 , that is likely to react with Sr in a way similar to how Cl_2 reacts with Sr. Justify your choice.
- (c) A graph of potential energy versus internuclear distance for two Cl atoms is given below. On the same graph, carefully sketch a curve that corresponds to potential energy versus internuclear distance for two Br atoms.



- (d) In the box below, draw a complete Lewis electron-dot diagram for the C_2Cl_4 molecule.

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(e) Answer the following based on the diagram you drew above.

(i) What is the hybridization of the C atoms in C_2Cl_4 ?

(ii) What is the approximate chlorine-carbon-chlorine bond angle in C_2Cl_4 ?

(iii) Is the C_2Cl_4 molecule polar?

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the bond is covalent. (No explanation is required, however, the $\text{Cl} - \text{Cl}$ bond is covalent because the two Cl atoms have the same electronegativity and will form a nonpolar covalent bond.)

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets both of the criteria below:

- ☐ F_2 , Br_2 , I_2 , or At_2 is written.
- ☐ the element chosen is in the same group or family as Cl (or Cl_2).

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The sketched curve meets both of the criteria below:

- ☐ The curve has its minimum value to the right of where the given curve has its minimum value (because the length of the $Br - Br$ bond is greater than that of the $Cl - Cl$ bond).
- ☐ The minimum value of the curve is higher than (i.e., above) the minimum value of the given curve (because the strength of the $Br - Br$ bond is less than that of the $Cl - Cl$ bond).

Note: Only the sketched curve is scored; parenthetical explanations are not scored.

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. Bonds can be drawn using lines or dots.



0	1	2
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The response meets both of the criteria below:

- ☐ The diagram has two carbon atoms connected to each other and two chlorine atoms connected to each of the carbon atoms.
- ☐ There is a double bond between the two carbon atoms, and there are no nonbonding pairs on any of the atoms.

Part (e)(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the hybridization is sp^2 . (No explanation is required, however the C atoms in C_2Cl_4 are surrounded by three electron domains, a double bond and two single bonds. This situation corresponds to sp^2 hybridization.)

Part (e)(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. (The actual bond angle is approximately 113° .)



0	1
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The response indicates that the bond angle is approximately 120° (any value from 110° to 125° is acceptable).

Part (e)(iii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the molecule is not polar. (No explanation is required; however, the molecule is not polar because the dipoles formed by unequal sharing of electrons in the C — Cl bonds cancel due to the symmetry of the molecule.)

57. Answer the following questions using principles of chemical bonding and molecular structure.

(a) Consider the carbon dioxide molecule, CO_2 , and the carbonate ion, CO_3^{2-} .

(i) Draw the complete Lewis electron-dot structure for each species.

(ii) Account for the fact that the carbon-oxygen bond length in CO_3^{2-} is greater than the carbon-oxygen bond length in CO_2 .

(b) Consider the molecules CF_4 and SF_4 .

(i) Draw the complete Lewis electron-dot structure for each molecule.

(ii) In terms of molecular geometry, account for the fact that the CF_4 molecule is nonpolar, whereas the SF_4 molecule is polar.

Part A (i)

1 point is earned for each Lewis electron-dot structure

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Indication of lone pairs of electrons are required on each structure

Resonance forms of CO₃²⁻ are not required



0	1	2
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The student response earns two of the following points:

1 point is earned for each Lewis electron-dot structure

Indication of lone pairs of electrons are required on each structure

Resonance forms of CO₃²⁻ are not required

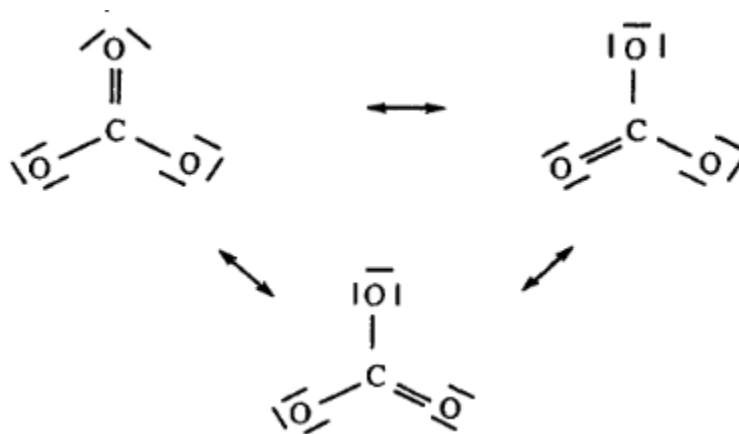
**Part A (ii)**

1st point is earned for indicating double bonds are present in CO₂ OR resonance occurs in CO₃²⁻.

2nd point earned for BOTH of the below AND indicating the relative lengths of the bond type

In CO₂, the C-O interactions are double bonds, OR, in CO₃²⁻ the C-O interactions are resonance forms (or figures below.)

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The carbon-oxygen bond length is greater in the resonance forms than in the double bonds



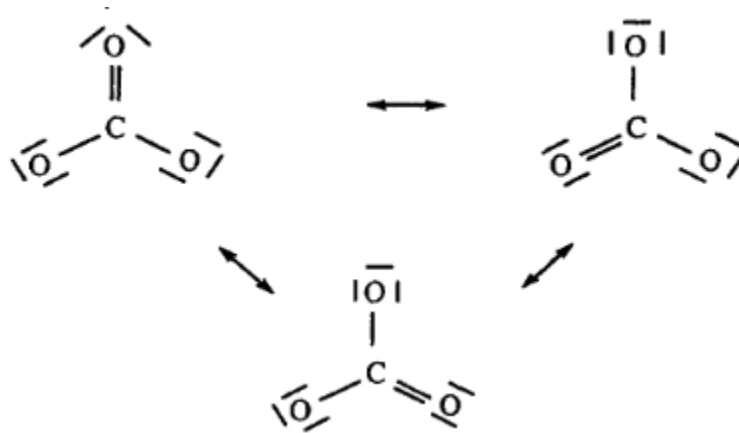
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The student response earns two of the following points:

1st point is earned for indicating double bonds are present in CO₂ OR resonance occurs in CO₃²⁻

2nd point earned for BOTH of the below AND indicating the relative lengths of the bond type

In CO₂, the C-O interactions are double bonds, OR, in CO₃²⁻ the C-O interactions are resonance forms (or figures below.)



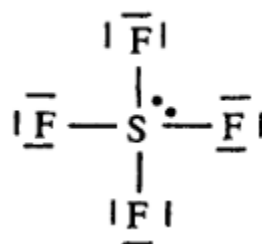
The carbon-oxygen bond length is greater in the resonance forms than in the double bonds

Part B (i)

One point earned for each Lewis electron-dot structure

Lone pairs of electrons are required on each structure

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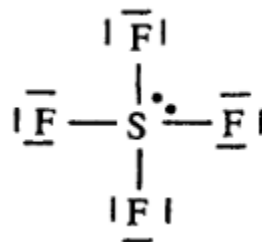


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The student response earns two of the following points:

One point earned for each Lewis electron-dot structure

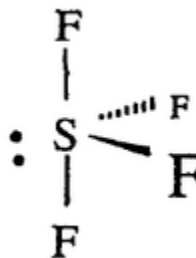
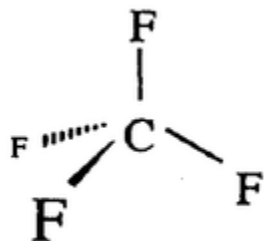
Lone pairs of electrons are required on each structure

**Part B (ii)**

One point earned for each molecule for proper geometry and explanation

CF₄ has a tetrahedral geometry, so the bond dipoles cancel, leading to a nonpolar molecule.

With five pairs of electrons around the central S atom, SF₄ exhibits a trigonal bipyramidal electronic geometry, with the lone pair of electrons. In this configuration, the bond dipoles do not cancel, and the molecule is polar.



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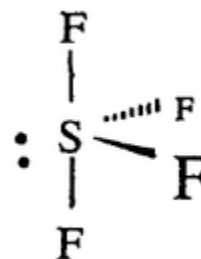
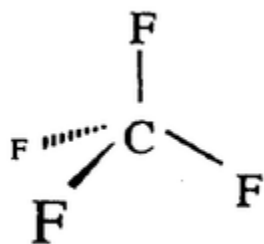
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The student response earns two of the following points:

One point earned for each molecule for proper geometry and explanation

CF₄ has a tetrahedral geometry, so the bond dipoles cancel, leading to a nonpolar molecule.

With five pairs of electrons around the central S atom, SF₄ exhibits a trigonal bipyramidal electronic geometry, with the lone pair of electrons. In this configuration, the bond dipoles do not cancel, and the molecule is polar.



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58.

Nonmetal	C	N	O	Ne	Si	P	S	Ar
Formula of Compound	CF ₄	NF ₃	OF ₂	No compound	SiF ₄	PF ₃	SF ₂	No compound

Some binary compounds that form between fluorine and various nonmetals are listed in the table above. A student examines the data in the table and poses the following hypothesis: the number of F atoms that will bond to a nonmetal is always equal to 8 minus the number of valence electrons in the nonmetal atom.

- a. Based on the student's hypothesis, what should be the formula of the compound that forms between chlorine and fluorine?
- b. In an attempt to verify the hypothesis, the student researches the fluoride compounds of the other halogens and finds the formula ClF₃. In the box below, draw a complete Lewis electron-dot diagram for a molecule of ClF₃.



- c. Two possible geometric shapes for the ClF₃ molecule are trigonal planar and T-shaped. The student does some research and learns that the molecule has a dipole moment. Which of the two shapes is consistent with the fact that the ClF₃ molecule has a dipole moment? Justify your answer in terms of bond polarity and molecular structure.

In an attempt to resolve the existence of the ClF₃ molecule with the hypothesis stated above, the student researches the compounds that form between halogens and fluorine, and assembles the following list.

Halogen	Formula(s)
F	F ₂
Cl	
Br	BrF, BrF ₃ , BrF ₅
I	IF, IF ₃ , IF ₅ , IF ₇

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- d. Based on concepts of atomic structure and periodicity, propose a modification to the student's previous hypothesis to account for the compounds that form between halogens and fluorine.

Part A

1 point is earned for the correct formula.

ClF



0

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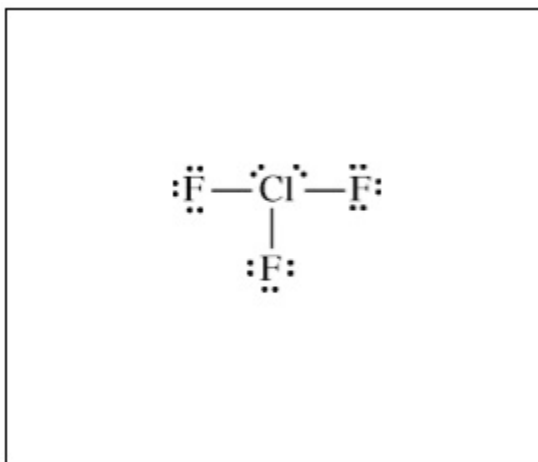
The student response earns one of the following points:

1 point is earned for the correct formula.

ClF

Part B

1 point is earned for a central Cl atom surrounded by three bonding pairs with F atoms and two nonbonding (lone) pairs of electrons. F atoms must have three nonbonding pairs each. Electron pairs can be depicted as dots or line segments.



See diagram above.



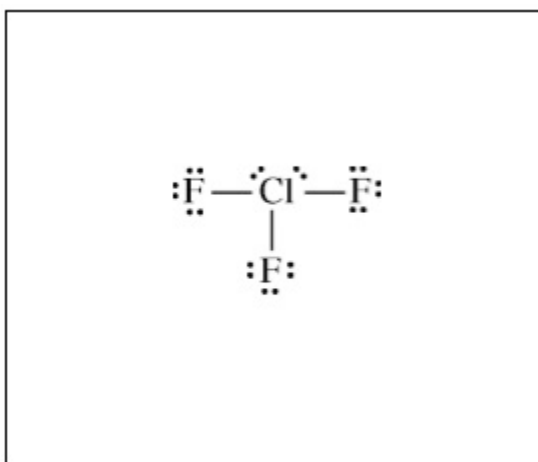
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The student response earns one of the following points:

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1 point is earned for a central Cl atom surrounded by three bonding pairs with F atoms and two nonbonding (lone) pairs of electrons. F atoms must have three nonbonding pairs each. Electron pairs can be depicted as dots or line segments.



See diagram above.

Part C

1 point is earned for indicating that the molecule is T-shaped with an acceptable explanation.

The molecule is T-shaped

because a T-shaped structure is asymmetric with dipoles that do not cancel out, but produce a net dipole (i.e., a polar molecule).

OR

because, if the molecule had a trigonal planar structure, the molecule would be symmetric with dipoles that cancel out and produce a net dipole of zero (i.e., a nonpolar molecule), which is not consistent with the observation that the ClF₃ molecule does have a dipole moment.



0	1
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The student response earns one of the following points:

1 point is earned for indicating that the molecule is T-shaped with an acceptable explanation.

The molecule is T-shaped

because a T-shaped structure is asymmetric with dipoles that do not cancel out, but produce a net dipole (i.e., a polar molecule).

OR

because, if the molecule had a trigonal planar structure, the molecule would be symmetric with dipoles that cancel out and

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produce a net dipole of zero (i.e., a nonpolar molecule), which is not consistent with the observation that the ClF_3 molecule does have a dipole moment.

Part D

1 point is earned for an acceptably modified hypothesis that addresses both atomic structure and periodicity.

An acceptable hypothesis (descriptive or formulaic) must include the following ideas:

- Atomic Structure: e.g., odd number of F atoms
- Periodicity: e.g., as the atomic number of the central halogen atom increases, the number of F atoms increases



0	1
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The student response earns one of the following points:

1 point is earned for an acceptably modified hypothesis that addresses both atomic structure and periodicity.

An acceptable hypothesis (descriptive or formulaic) must include the following ideas:

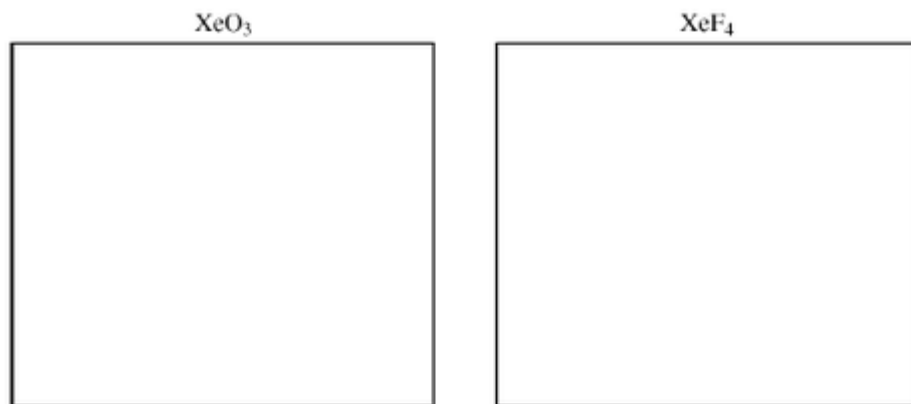
- Atomic Structure: e.g., odd number of F atoms
- Periodicity: e.g., as the atomic number of the central halogen atom increases, the number of F atoms increases

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59. Using principles of atomic and molecular structure and the information in the table below, answer the following questions about atomic fluorine, oxygen, and xenon, as well as some of their compounds.

Atom	First Ionization Energy (kJ mol ⁻¹)
F	1,681.0
O	1,313.9
Xe	?

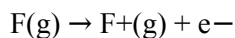
- (a) Write the equation for the ionization of atomic fluorine that requires 1,681.0 kJ mol⁻¹.
- (b) Account for the fact that the first ionization energy of atomic fluorine is greater than that of atomic oxygen. (You must discuss both atoms in your response.)
- (c) Predict whether the first ionization energy of atomic xenon is greater than, less than, or equal to the first ionization energy of atomic fluorine. Justify your prediction.
- (d) Xenon can react with oxygen and fluorine to form compounds such as XeO₃ and XeF₄. In the boxes provided, draw the complete Lewis electron-dot diagram for each of the molecules represented below.



- (e) On the basis of the Lewis electron-dot diagrams you drew for part (d), predict the following:
- The geometric shape of the XeO₃ molecule
 - The hybridization of the valence orbitals of xenon in XeF₄
- (f) Predict whether the XeO₃ molecule is polar or nonpolar. Justify your prediction.

Part A

1 point is earned for the correct equation. (Phase designations are not required.)

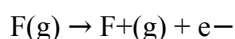
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✓

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The student response earns one of the following points:

1 point is earned for the correct equation. (Phase designations are not required.)

**Part B**

1 point is earned for recognizing that the effective nuclear charge of F is greater than that of O.

In both cases the electron removed is from the same energy level ($2p$), but fluorine has a greater effective nuclear charge due to one more proton in its nucleus (the electrons are held more tightly and thus take more energy to remove).

✓

0	1
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The student response earns one of the following points:

1 point is earned for recognizing that the effective nuclear charge of F is greater than that of O.

In both cases the electron removed is from the same energy level ($2p$), but fluorine has a greater effective nuclear charge due to one more proton in its nucleus (the electrons are held more tightly and thus take more energy to remove).

Part C

1 point is earned for a prediction based on size and/or energy level.

The first ionization energy of Xe should be less than the first ionization energy of F. To ionize the F atom, an electron is removed from a $2p$ orbital. To ionize the Xe atom, an electron must be removed from a $5p$ orbital. The $5p$ is a higher energy level and is farther from the nucleus than $2p$, hence it takes less energy to remove an electron from Xe.

✓

0	1
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The student response earns one of the following points:

1 point is earned for a prediction based on size and/or energy level.

The first ionization energy of Xe should be less than the first ionization energy of F. To ionize the F atom, an electron is removed from a $2p$ orbital. To ionize the Xe atom, an electron must be removed from a $5p$ orbital. The $5p$ is a higher

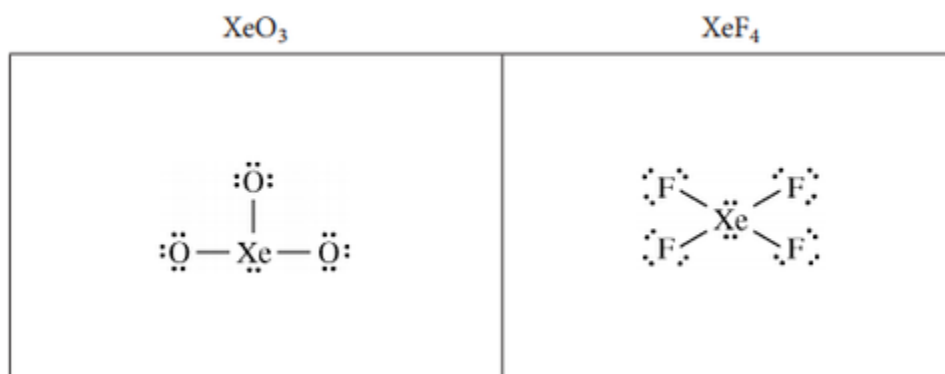
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energy level and is farther from the nucleus than $2p$, hence it takes less energy to remove an electron from Xe.

Part D

1 point is earned for each correct Lewis electron-dot diagram.

Omission of lone pairs of electrons on the O or F atoms results in a one-time 1-point deduction.



✓

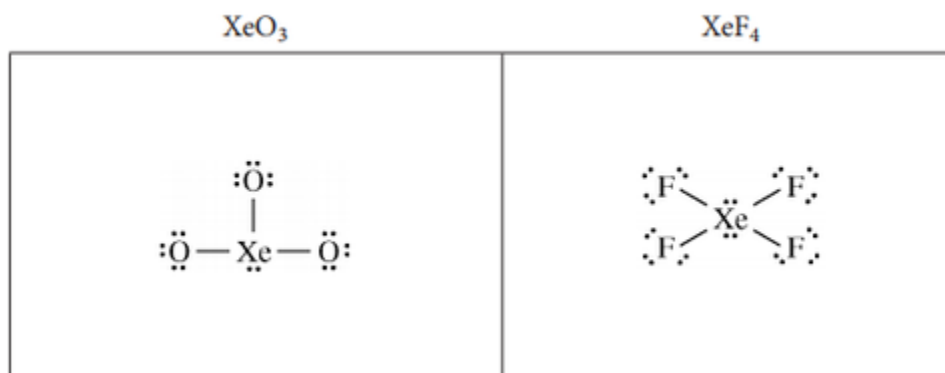
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The student response earns one of the following points:

1 point is earned for each correct Lewis electron-dot diagram.

Omission of lone pairs of electrons on the O or F atoms results in a one-time 1-point deduction.

**Part E (i)**

1 point is earned for a shape that is consistent with the Lewis electron-dot diagram.

Trigonal pyramidal

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✓

0	1
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The student response earns one of the following points:

1 point is earned for a shape that is consistent with the Lewis electron-dot diagram.

Trigonal pyramidal

Part E (ii)

1 point is earned for the hybridization consistent with the Lewis electron-dot diagram.

sp^3d^2

✓

0	1
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The student response earns one of the following points:

1 point is earned for the hybridization consistent with the Lewis electron-dot diagram.

sp^3d^2

Part F

1 point is earned for the answer that is consistent with the shape indicated in part (e)(i).

1 point is earned for an explanation correctly related to the shape in part (e)(i).

The XeO_3 molecule would be polar because it contains three polar Xe–O bonds that are asymmetrically arranged around the central Xe atom (i.e., the bond dipoles do not cancel but add to a net molecular dipole with the Xe atom at the positive end).

✓

0	1	2
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The student response earns two of the following points:

1 point is earned for the answer that is consistent with the shape indicated in part (e)(i).

1 point is earned for an explanation correctly related to the shape in part (e)(i).

The XeO_3 molecule would be polar because it contains three polar Xe–O bonds that are asymmetrically arranged around

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the central Xe atom (i.e., the bond dipoles do not cancel but add to a net molecular dipole with the Xe atom at the positive end).